

layers of the mosaic grid,
with the fluid between them

violet: the fluid between the
layers (dark matter); it fills
every gap and pools around
the mass

layers above the mass:
pulled DOWN toward it

layer level with the mass:
pulled INWARD sideways

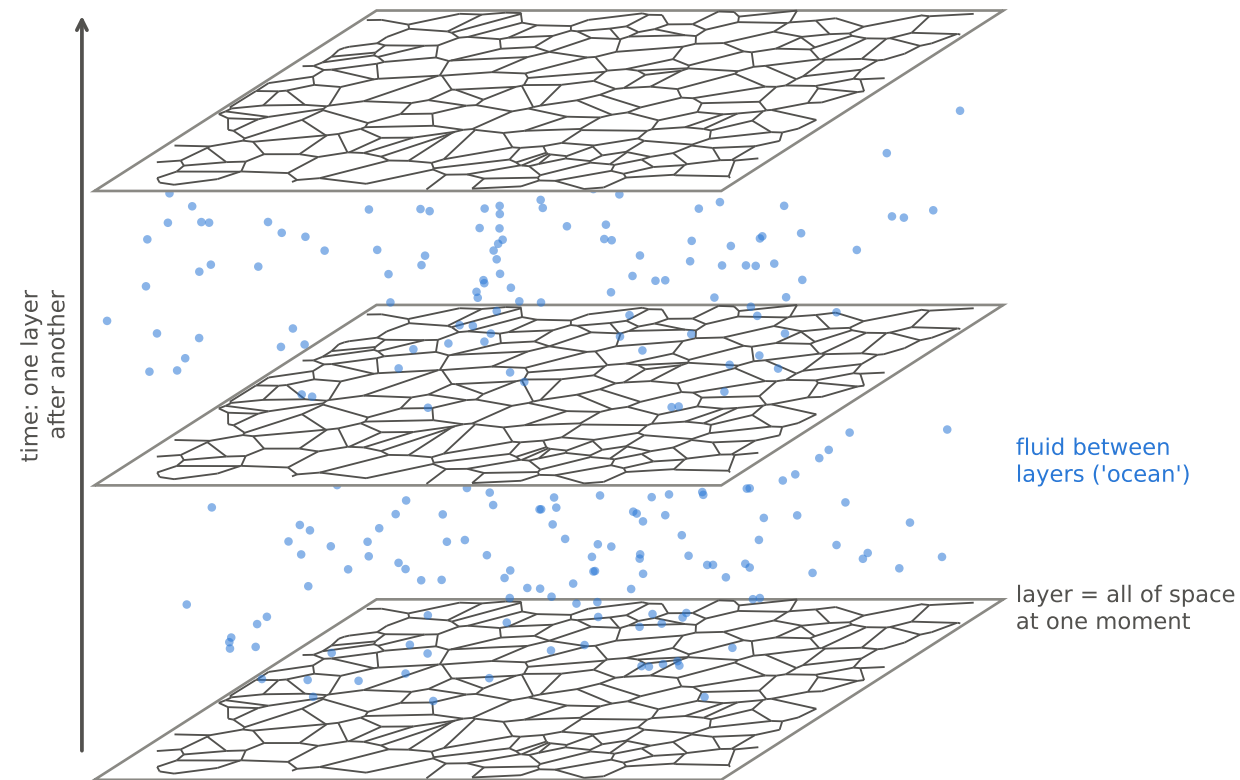
layers below the mass:
pulled UP toward it

green: an irregular orbit inside the fluid pool;
it never closes, tracing a rosette

gold: an elliptical orbit (tilted); it closes on itself
pale blue: light bending toward the mass;
dotted = straight lines

$\nabla^2 \phi = 4\pi G \rho$, draw-in $\mathbf{g} = -\nabla \phi = -GM\hat{r}/r^2$; orbit $\ddot{\mathbf{x}} = \mathbf{g}$; light $n = 1 - 2\phi/c^2$, bend $\alpha = 4GM/bc^2$; tick rate $d\tau/dt \approx 1 + \phi/c^2$

Figure 1. Anatomy of the grid



WHAT THE PIECES ARE

Links: carry tension, like the strings of a stretched net. Their pull is what we feel as gravity (Fig. 2-3).

Cells: random, not a crystal, so no direction is special (a regular grid would show its axes; we measured $\sim 20\%$ for a cube lattice).

Layers: one per moment, like the pages of a flip-book. Relativity's 'preferred slicing' in Khronon theory is this stack.

Fluid: fills the gaps between pages. It is pulled by gravity but does not collide or shine: the dark matter of the CMB, clusters, lensing.

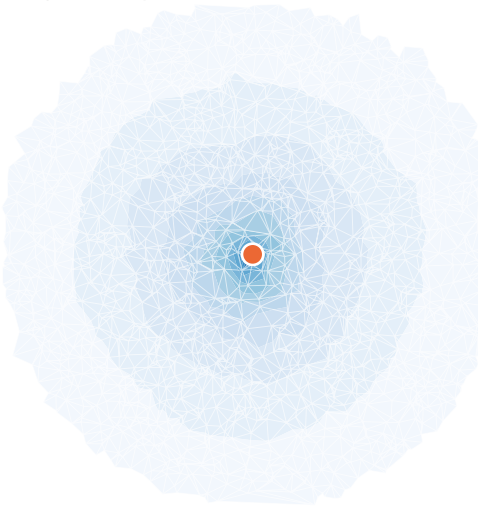
Cosmic tension: the net's overall stretch is the dark energy (Fig. 4).

picture (layout); cell geometry is a real random mosaic

Figure 2. Gravity is the grid's tension, shared out link by link

(a) A mass pulls on the net

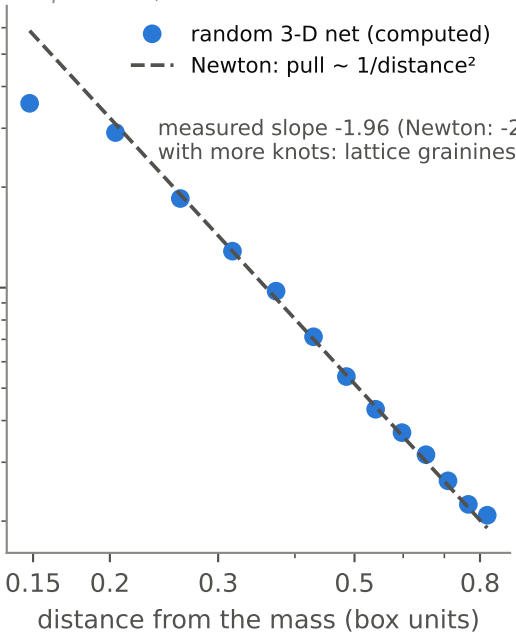
computed: equilibrium of a 2-D random network



a random net, tension only; the mass pulls one knot.
darker = pulled further. Every knot shares the load
with its neighbours, so the dip spreads out smoothly.

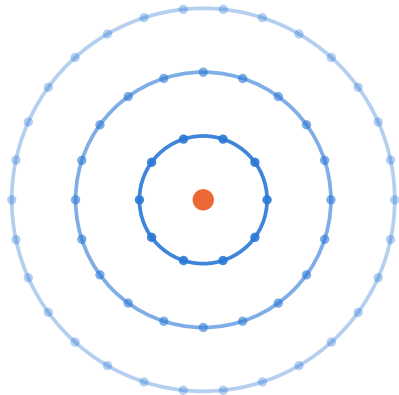
(b) The pull falls as $1/r^2$

computed: 24,000-knot random 3-D network



(c) Why $1/r^2$

picture



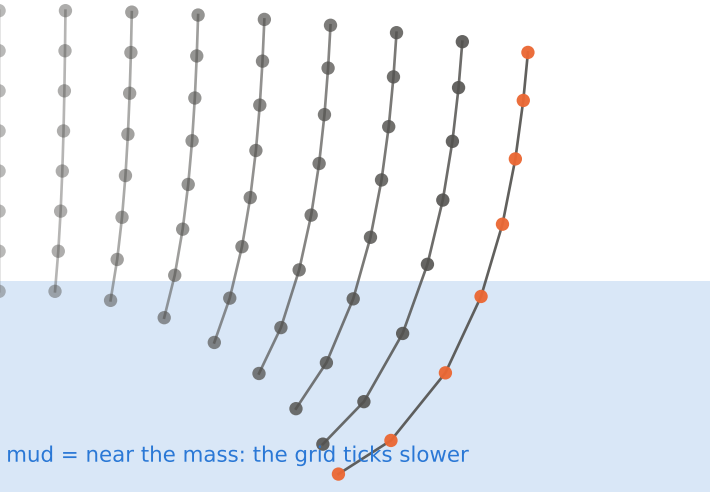
the same pull is shared by every link crossing
a shell; a shell twice as far has 4x as many,
so each carries 1/4: the inverse-square law.

Figure 3. What gravity actually is: the grid ticks slower near mass, and paths bend toward slow ticking

(a) Falling = turning toward slower ticks

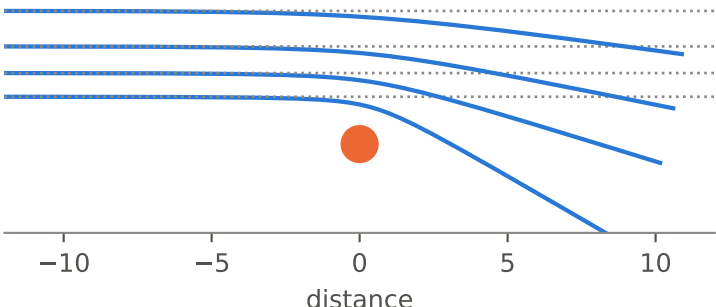
picture (computed toy: step length = local tick rate)

a marching band whose lower ranks reach mud
each row turns toward the slow side:
no rope pulls it; the path itself bends



(b) Light takes the same bent path

computed: ray tracing through the tick-rate field (exaggerated)



dotted: straight lines. Rays passing closer bend more.
The Sun bends starlight by 1.75" (measured 1919 onward);
here the mass is exaggerated $\sim 10^5$ times.

Figure 4. Dark energy is the grid's own tension, and it 'gives' when the stretching speeds up

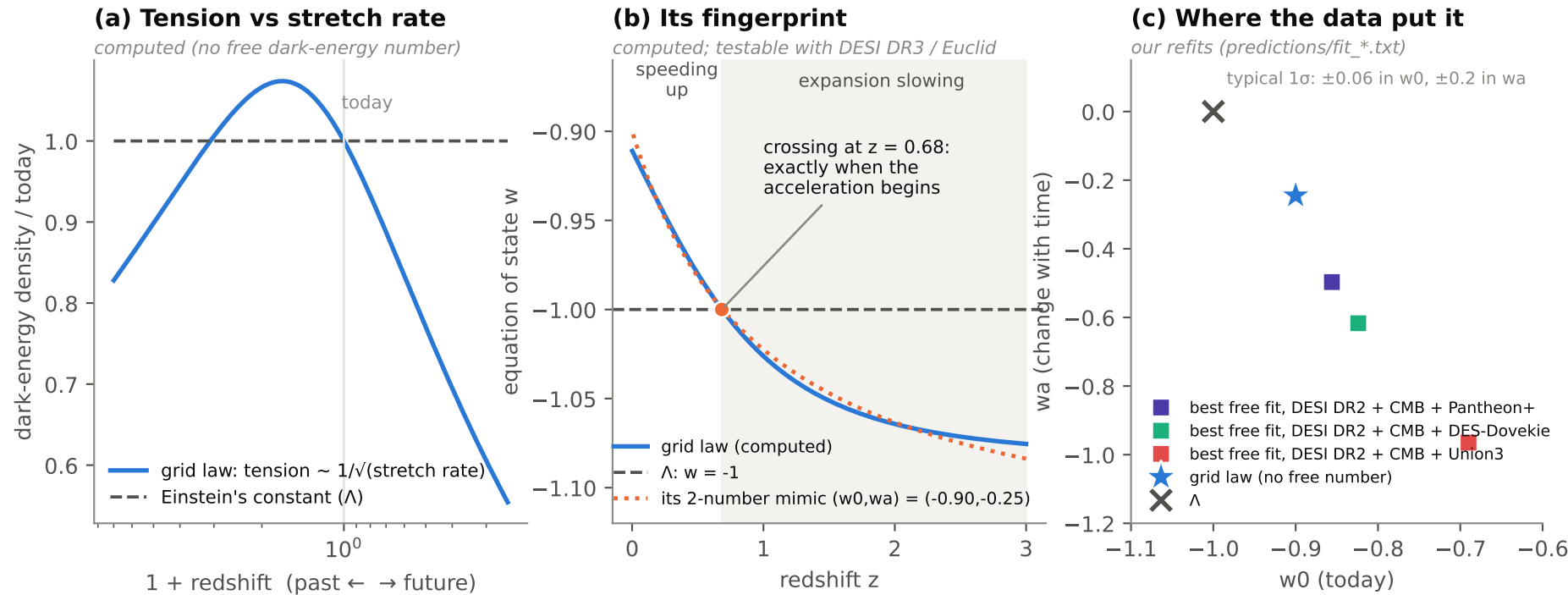
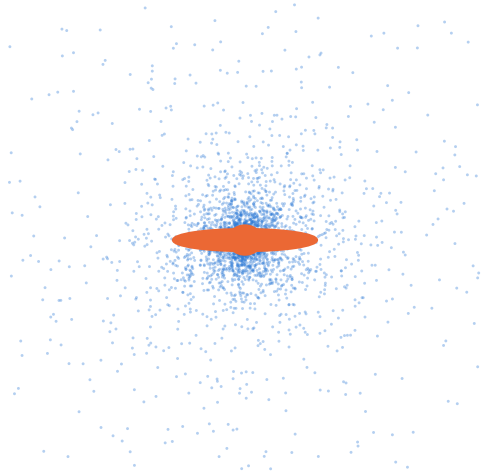


Figure 5. The fluid between the layers: pulled by gravity, invisible, and it passes through itself

(a) A galaxy sits in a pool of fluid

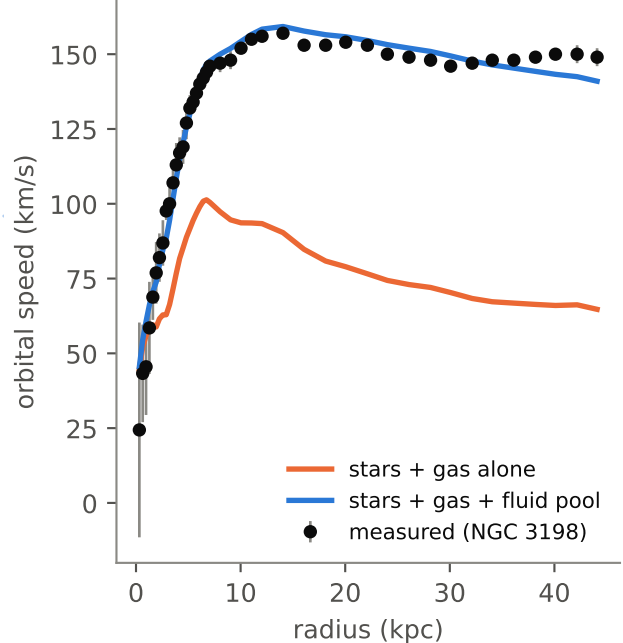
picture (profile drawn from a CDM-like halo)



orange: stars and gas. blue: the fluid it has gathered (5-6x more mass). The pool holds the outer stars in orbit (flat rotation curves).

(b) Why we need the fluid

computed: SPARC data, Newtonian + cored halo fit



(c) Two clusters after a collision

picture of the Bullet Cluster observation

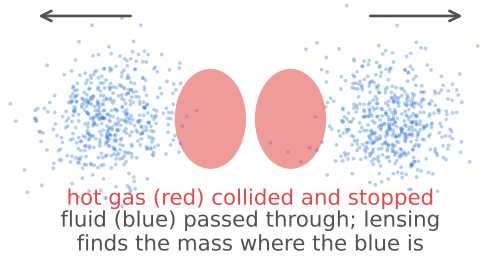


Figure 6. The history of the universe in the grid picture, with what has been tested

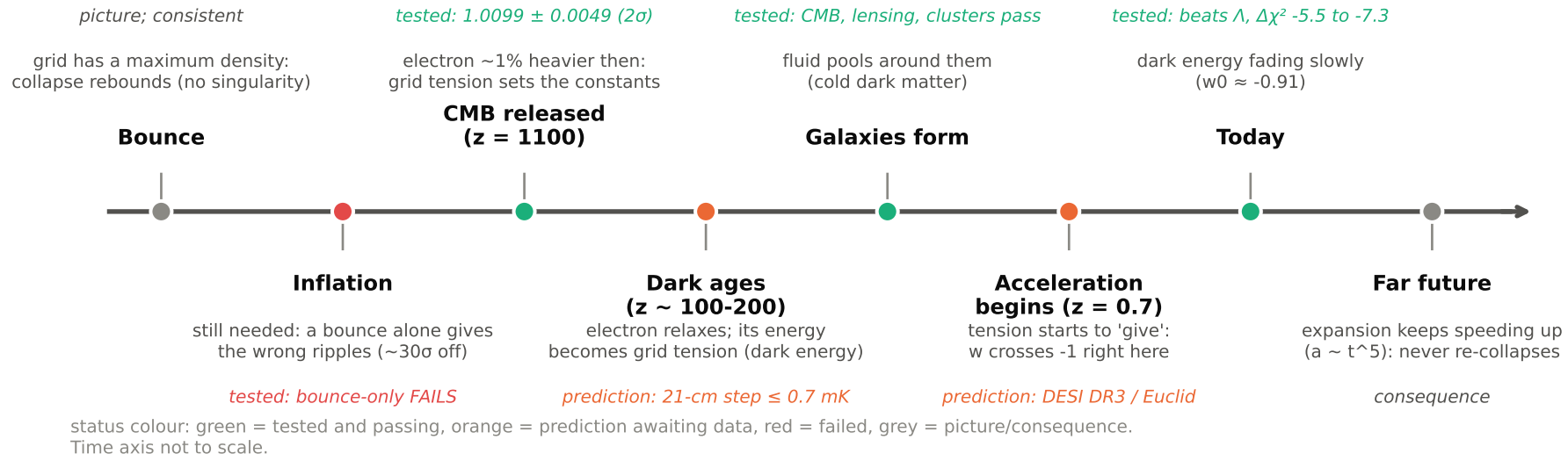


Figure 7. Matter as patterns in the grid: the electron's two handed halves live on two layers

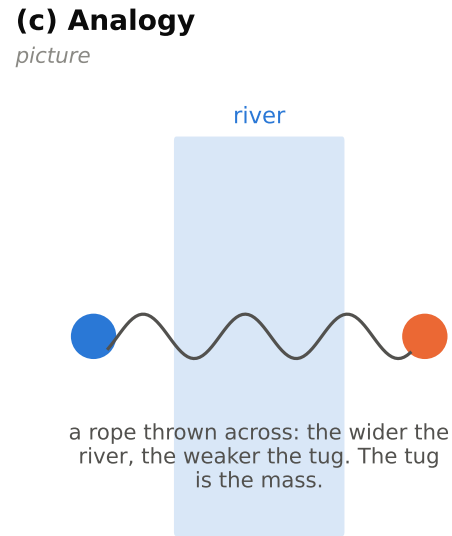
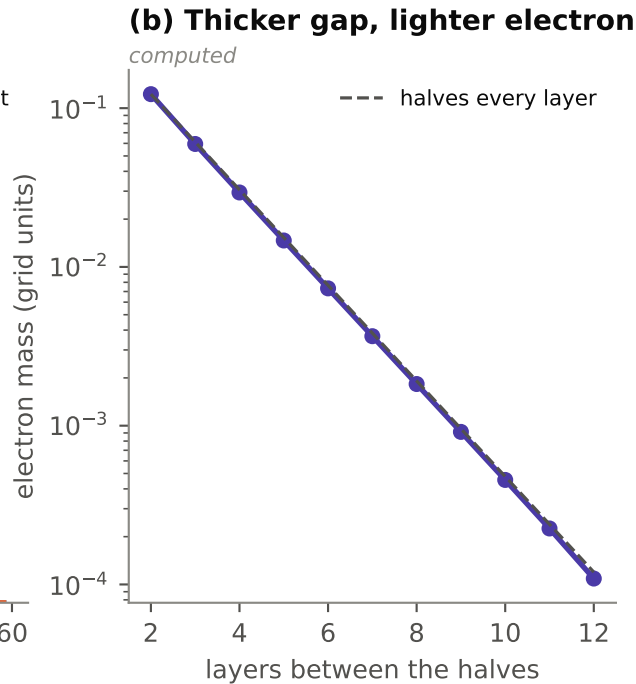
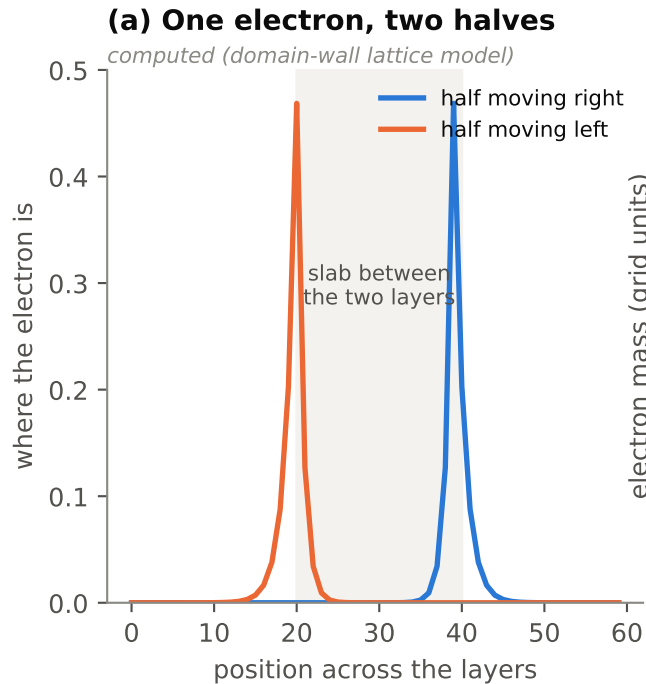
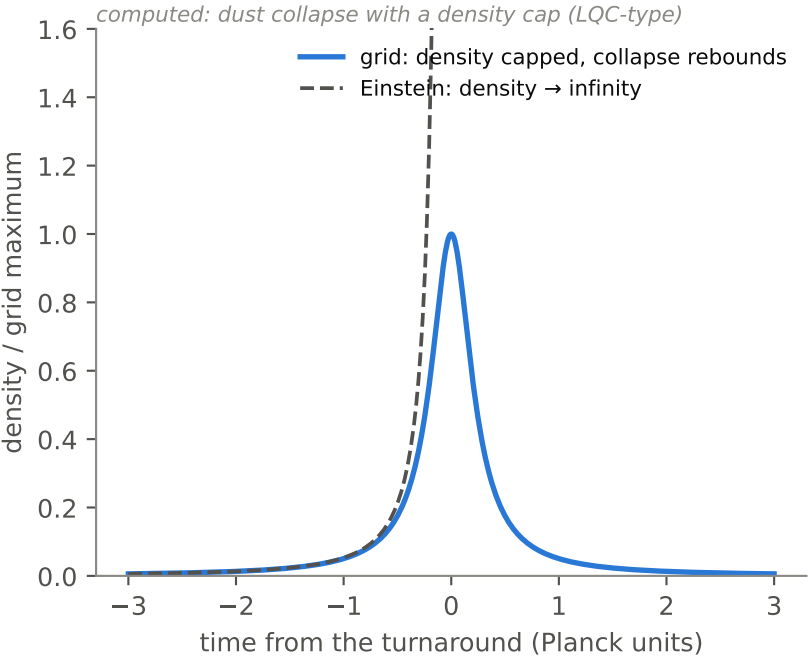


Figure 8. Black holes: the grid cannot be compressed without limit

(a) The core of a collapsing star



(b) Analogy

picture

a full lift: people cannot squeeze any closer,
so the push turns into a rebound

The diagram shows a 5x5 grid of circles. The circles in the four corners (positions (1,1), (1,5), (5,1), and (5,5) assuming top-left is (1,1)) are orange. All other circles in the grid are blue. This represents a checkerboard pattern with orange circles at the corners.

result: a 'Planck star' at the centre, hidden by the horizon;
no observable difference for $>1e26$ years

Figure 9. Was the electron heavier when the CMB was released?

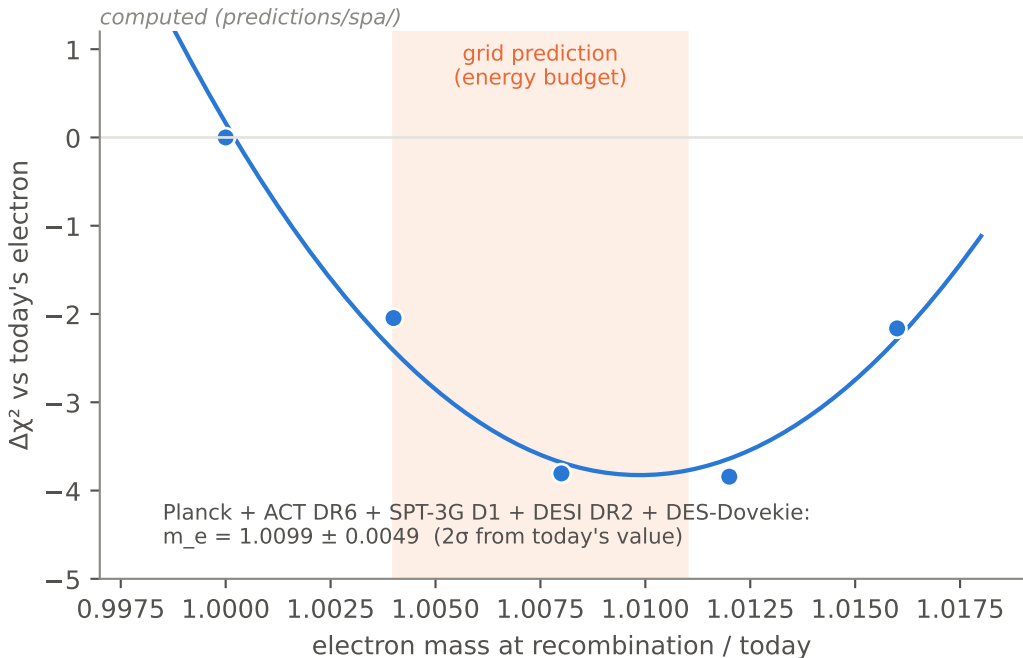
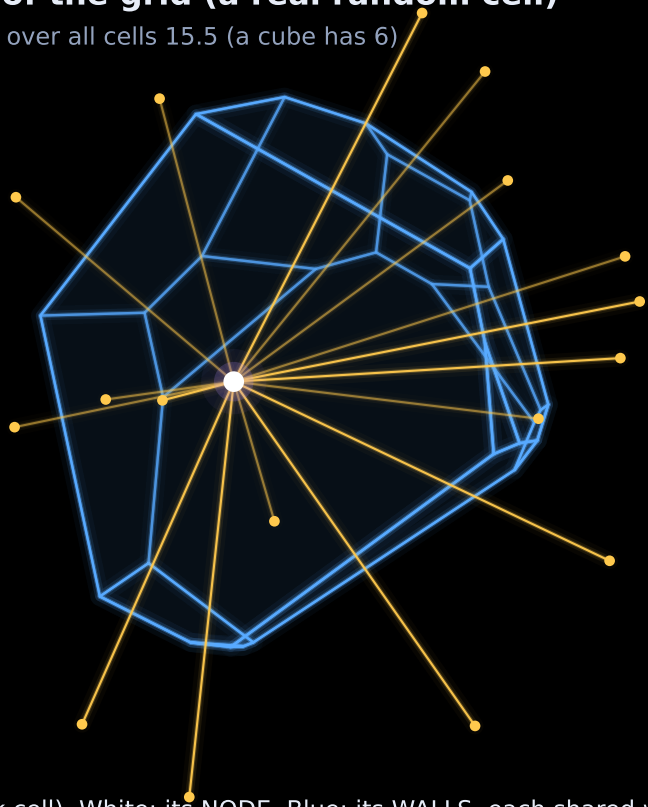


Figure 10. The grid's building block: a random cell, its links, and the size it must have

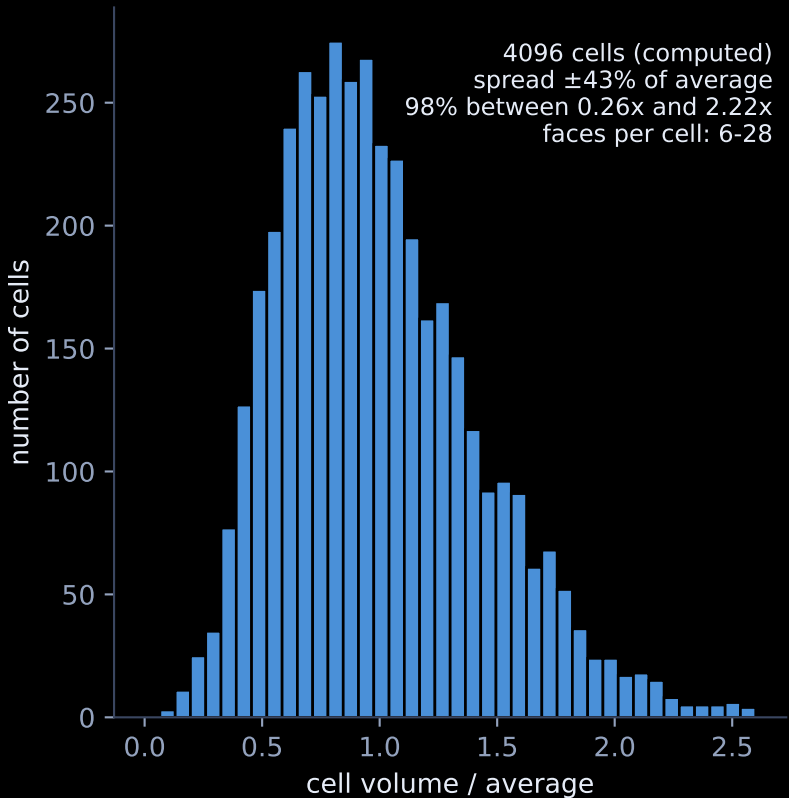
(a) One cell of the grid (a real random cell)

17 faces; average over all cells 15.5 (a cube has 6)

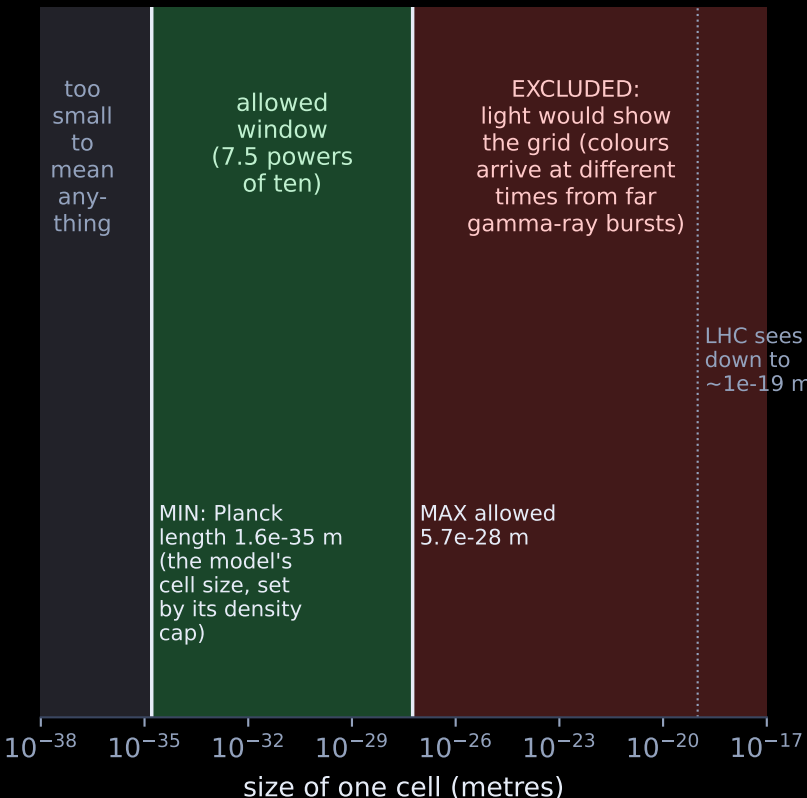


GRID CELL (Planck cell). White: its NODE. Blue: its WALLS, each shared with one neighbouring cell. Gold: its TENSION LINKS, one through each wall to the neighbouring node; a link's strength is set by its wall's area.

(b) Cells come in a spread of sizes



(c) How big can a cell be?



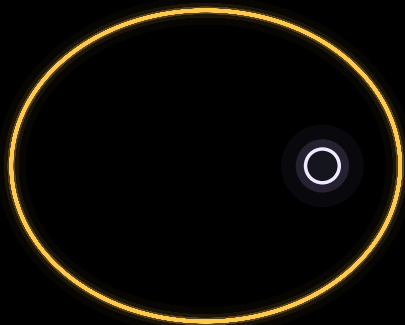
No cell needs a set size or shape: only the AVERAGE size matters, and randomness is what keeps every direction equal.

Equations: tension link weight $w_{ij} = A_{ij}/d_{ij}$ (wall area / link length); field on the grid $\sum_j w_{ij}(\phi_j - \phi_i) = 4\pi G m_i$

MAX cell size $\ell_{max} = \sqrt{12} \hbar c/E_{QG,2} = 5.7 \times 10^{-28} \text{ m}$ ($E_{QG,2} > 1.2 \times 10^{12} \text{ GeV}$, GRB 221009A); MIN $\ell_p = \sqrt{\hbar G/c^3} = 1.6 \times 10^{-35} \text{ m}$

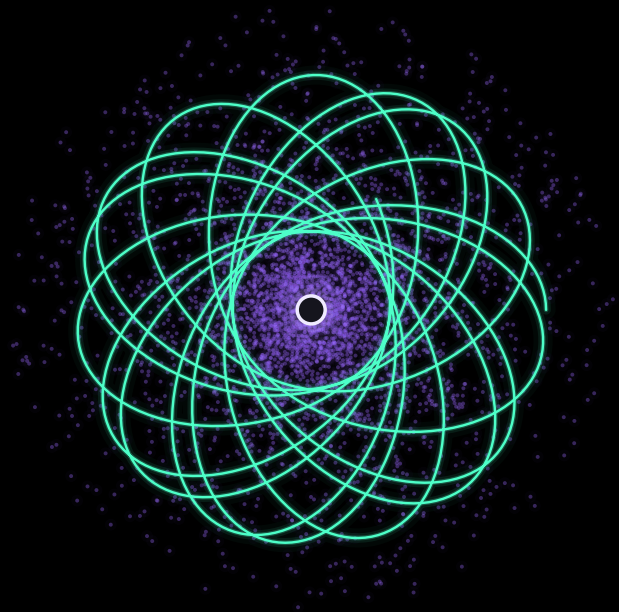
Figure 11. Three kinds of orbit in the pulled grid (computed; seen from above)

(a) Elliptical orbit around a lone mass



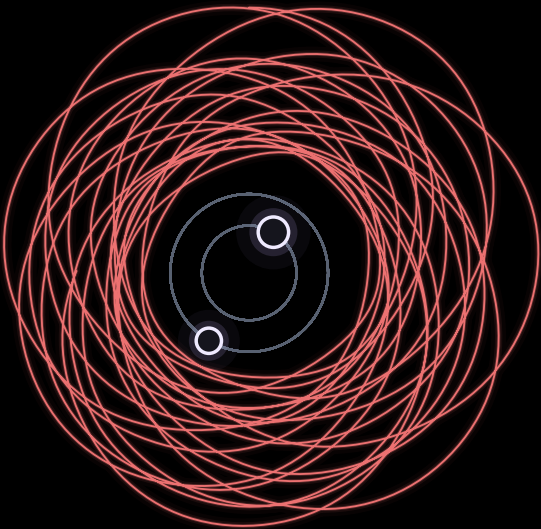
five laps drawn: they land exactly on top of each other. The mass sits at one focus; the body speeds up when close and slows when far (Kepler's laws).

(b) Irregular orbit inside the fluid pool



the pull grows as the body dips deeper into the fluid, so each lap swings round further: the orbit never closes and fills a ring (how stars move in galaxies).

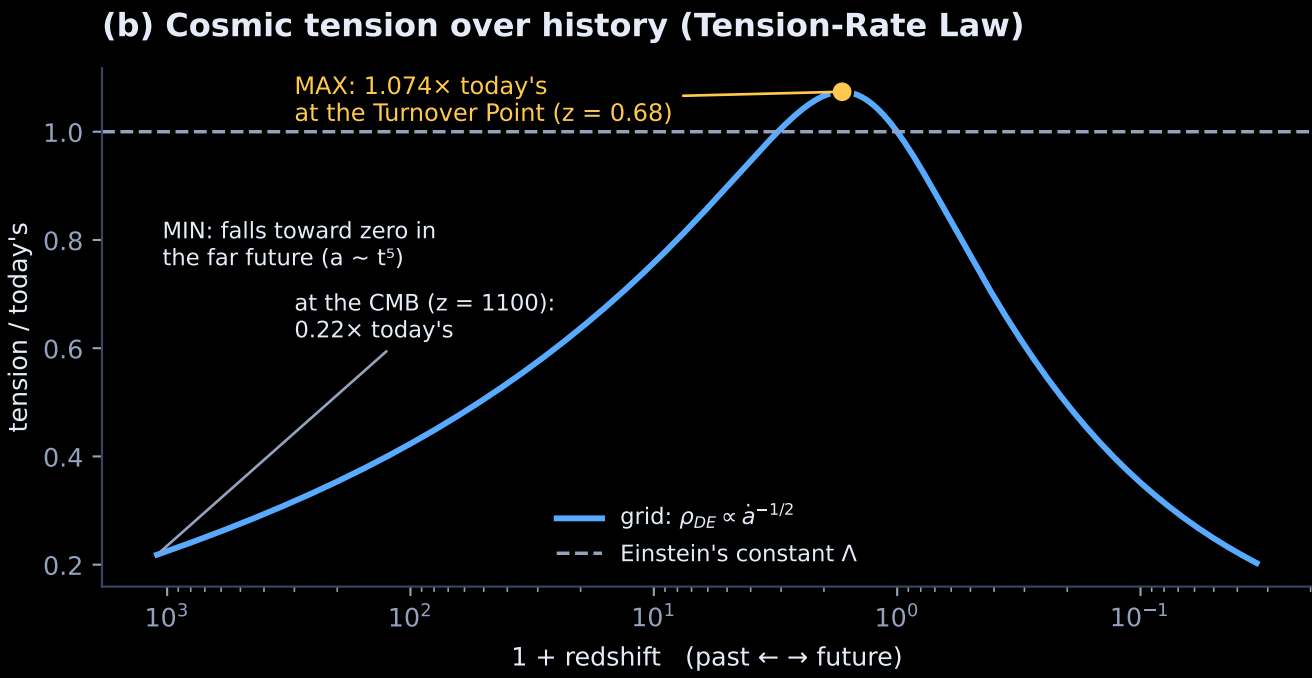
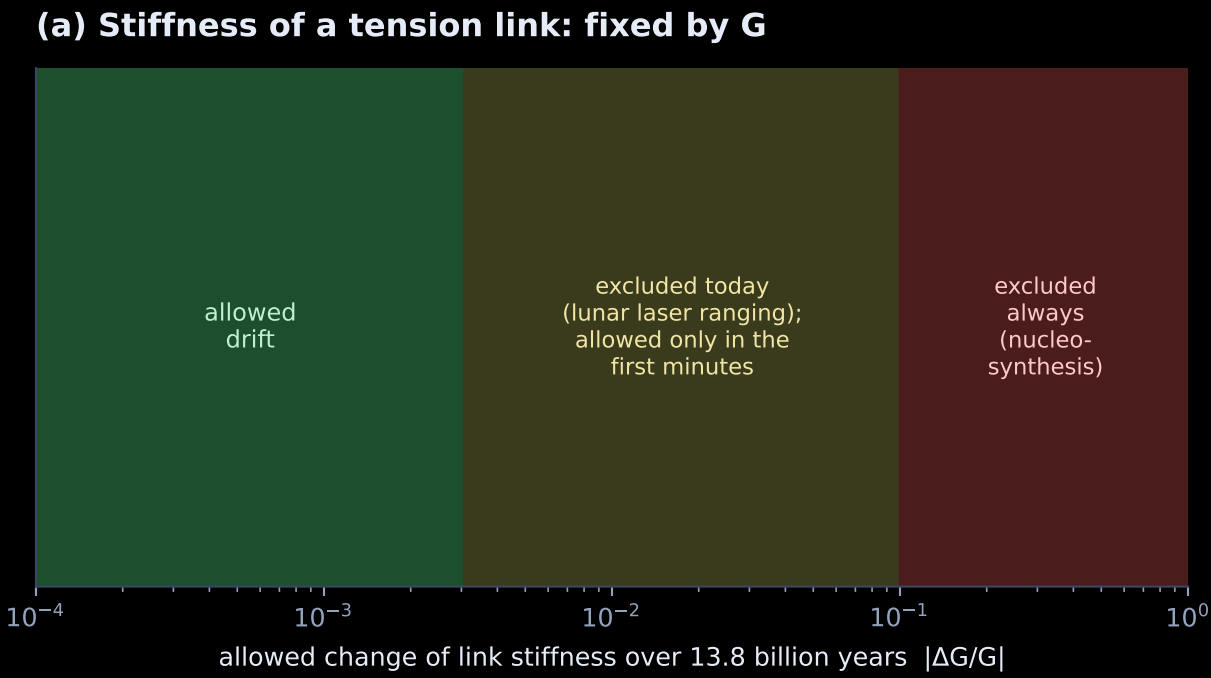
(c) Irregular orbit near two masses



two masses circle each other (grey); a light body is tugged by both and never repeats. Shift its start by one part in a million and the gap grows $\sim 1,629$ -fold.

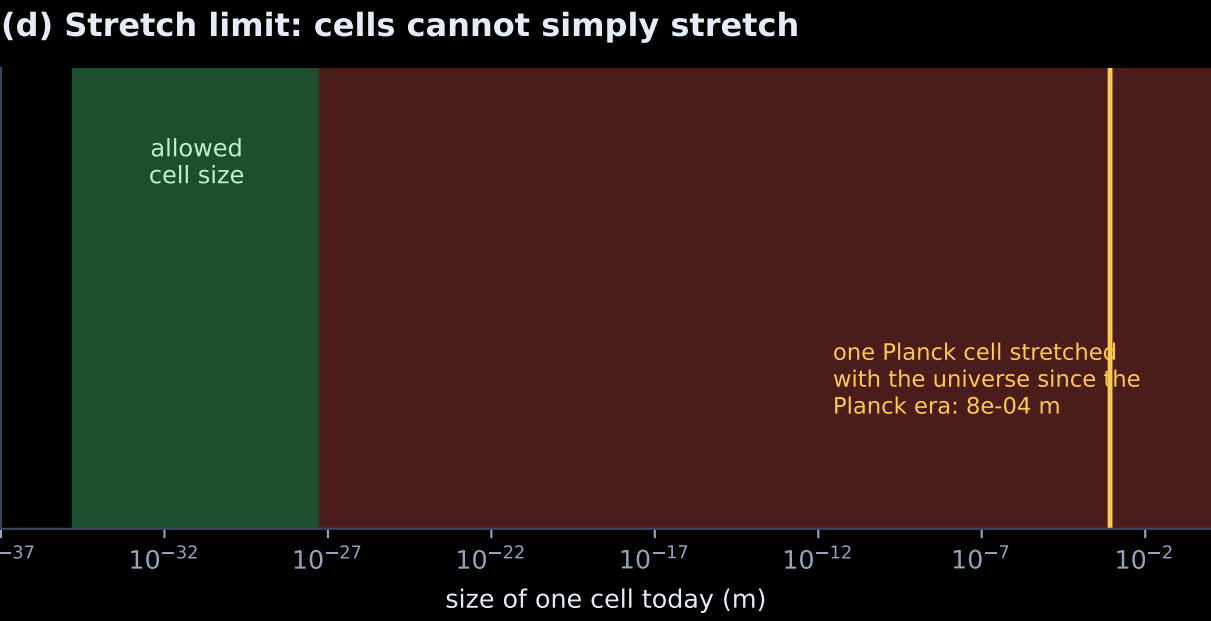
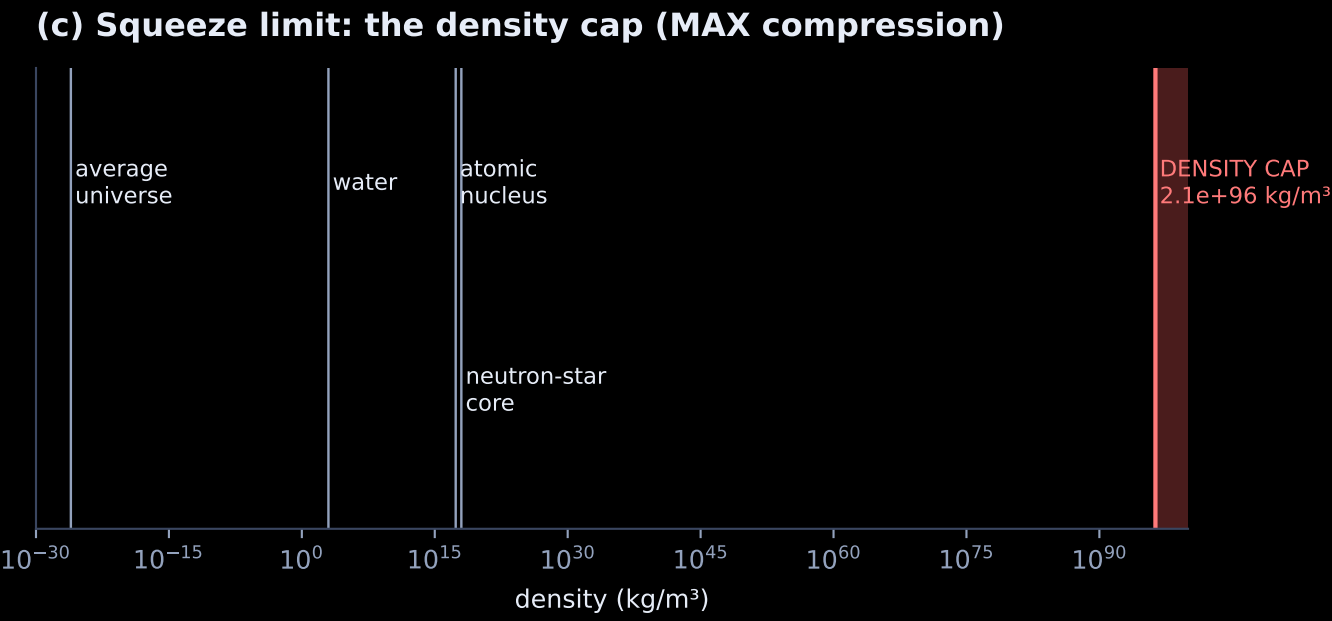
Equations: $\ddot{\mathbf{x}} = -\nabla\phi$. (a) $\phi = -GM/r$; $r = a(1 - e^2)/(1 + e\cos\theta)$, $T^2 = 4\pi^2 a^3/GM$. (b) $M(r) = M + M_h r^3/(r^2 + r_c^2)^{3/2}$, $\ddot{\mathbf{x}} = -GM(r)\hat{r}/r^2$. (c) $\phi = -\sum_k Gm_k/|\mathbf{x} - \mathbf{x}_k(t)|$; gap $\delta(t) \approx \delta_0 e^{\lambda t}$.

Figure 12. The grid's elasticity and its limits: how stiff, how tense, how far it can be squeezed and stretched



link tension scale $F = c^4/G = 1.21e + 44$ N (Planck force); stiffness of space $c^4/8\pi G = 4.8e + 42$ N
drift: $\dot{G}/G = (7.1 \pm 7.6) \times 10^{-14} \text{ yr}^{-1} \Rightarrow |\Delta G/G| < 3.1e - 03$ (2 σ) over the age of the universe

today: $\rho_{DEC^2} = 5.3e - 10$ J/m³ (= tension per area per length, Pa);
law: $d \ln \rho_{DE} / d \ln a = \frac{1}{2} q$, $q = -\ddot{a} a / \dot{a}^2$; MAX where $q = 0$ (the Turnover Point)



$\rho_{cap} \approx 0.41 \rho_P$, $\rho_P = c^5/\hbar G^2$; collapse obeys $H^2 = \frac{8\pi G}{3} \rho (1 - \rho/\rho_{cap})$:
at the cap the squeeze turns into a rebound (Bounce, Planck star).
Assumed from loop quantum cosmology, not measured.

stretch since the Planck era = $T_P/T_{CMB} = 5.2e + 31$; allowed stretch per cell $\leq \ell_{max}/\ell_P = 3.5e + 07$
 $\Rightarrow$ the grid must ADD cells as space expands (OPEN requirement of the model).